Spring 2013 / EC571

Instructor:Ajay Joshi

Exam 2

Date: April 10, 2013

Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	25	
2	25	
3	25	
4	25	
5	25	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours
2. The maximum grade is 100 points
3. Solve any 4 questions. If you solve all 5 questions, then the 4 best scores will be chosen to calculate the final grade.
4. Please write your name on all pages
5. The solutions should be written in the spaces provided on the test sheet
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number
7. This is an open book and open notes exam.
8. Material sharing is not allowed
9. Calculators are allowed
10. Boston University Academic Code of Conduct applies

Question 1

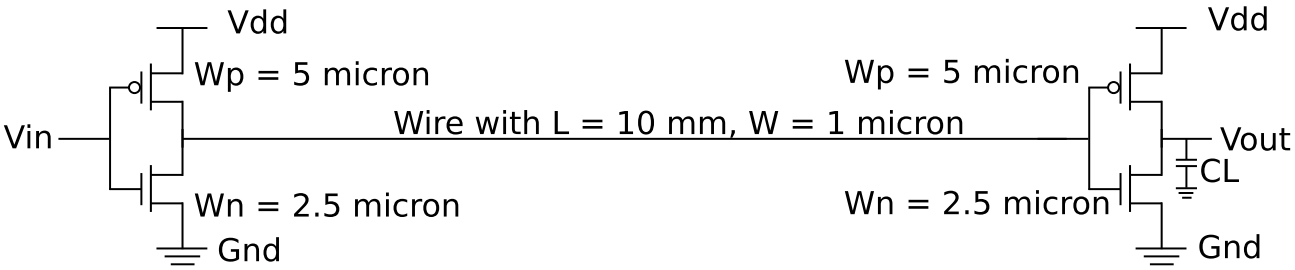


Figure 1

1. Draw the equivalent RC network. You can assume a distributed RC model with 'n' segments for the wire.

2. Write the Elmore delay equation for the design. Assume a step input at Vin.

3. Using the following technology parameters calculate the total Elmore delay from Vin to Vout when Vin switches from 0 V to 2.5 V (V_{dd} = 2.5 V, L = 0.25 micron $R_{eq}(NMOS)$ = 13 Kohm/sq, $R_{eq}(PMOS)$ = 26 Kohm/sq, C_g = 1.5 fF/micron, C_d = 2 fF/micron, C_{wire} = 0.2 fF/micron, R_{wire} = 100 kohm/micron, C_L = 50 fF), n = 10.

4. Calculate the total dynamic energy consumption of the circuit over 100 cycles. Assume 40 0 $\rightarrow$ 1 transitions and 40 1 $\rightarrow$ 0 transitions. Ignore the input capacitance of the first inverter.

Question 2

1. Design the following logic function using domino logic design: $f_1 = A \cdot B$.

2. Draw the equivalent RC network for the circuit.

3. Calculate the transistor widths such that the precharge operation happens in 250 psec and the evaluate operation happens in 250 psec. Assume the following technology parameters: $R_{sq(NMOS)}$ = 13 kohm/sq, $R_{sq(PMOS)}$ = 26 kohm/sq, $C_{g(NMOS)} = C_{g(PMOS)}$ = 1.5 fF/μm, $C_{d(NMOS)} = C_{d(PMOS)}$ = 2 fF/μm, C_L = 50 fF, and the minimum value for channel length and width of NMOS/PMOS is 300 nm and 450 nm, respectively.

Question 3

Draw the schematic for a positive-level triggered latch and a negative-level triggered latch. You can assume that both latches have been designed using transmission gates. Design a positive-edge triggered flip-flop using the two latches.

Question 4

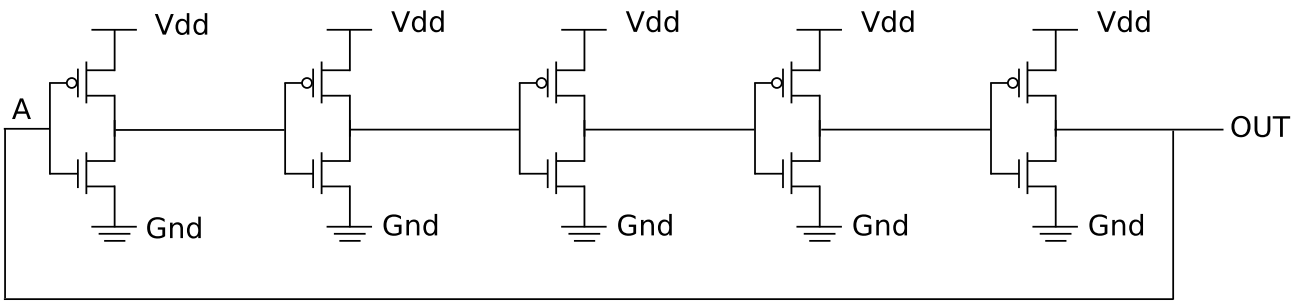


Figure 2

1. Figure 2 shows the schematic of a ring oscillator. Assume that at t = 0, a signal is applied (for a very short period of time) at node A, which triggers the first inverter and sets its output to Vdd. Draw the timing waveform of the signal at node OUT.

2. What would be the output waveform if the last inverter is removed from the circuit?

3. Assuming 5 inverters in the circuit, $R_{sq(NMOS)}$ = 13 kohm/sq, $R_{sq(PMOS)}$ = 33.5 kohm/sq, $C_{g(NMOS)}$ = $C_{g(PMOS)}$ = 1.5 fF/μm, $C_{d(NMOS)}$ = $C_{d(PMOS)}$ = 2 fF/μm, W_{NMOS} = 1.0 micron, L = 0.25 micron and W_{PMOS} = 2.5 micron, what is the time period of the signal at node OUT (in sub-problem 1). You can assume that the delay of the wire connecting the output of the last inverter to the first inverter is negligible.

Question 5

Design the function $f = \overline{A \cdot B + C \cdot D}$ using static complementary CMOS logic and pass transistor logic, such that it can operate at 2 GHz. For pass transistor logic you should use NMOS transistors. You can also assume that a minimum-sized inverter is used at the output of the pass transistor logic tree. Compare the area (in terms of the total transistor widths) and average energy consumed by the two designs. You can use $R_{sq(NMOS)}$ = 13 kohm/sq, $R_{sq(PMOS)}$ = 26 kohm/sq, $C_{g(NMOS)} = C_{g(PMOS)}$ = 1.5 fF/μm, $C_{d(NMOS)} = C_{d(PMOS)}$ = 2 fF/μm, C_L = 50 fF, and the minimum value for channel length and width of NMOS/PMOS is 300 nm and 450 nm, respectively.

Bonus question

Prof. Joshi is thinking of having a barbecue party at his place on Patriots day. Given that he is a vegetarian, suggest five food items that he can cook on a barbecue grill.